

Research Notes on Residual connection neural network

Residual connection neural network

$$a^{(l)} = \sigma(W^{(l)}a^{(l-1)} + b^{(l)})$$

>> Section 1

where ϕ can be any activation (e.g. ReLU) or normalization (e.g. LayerNorm) operation. This design reduces the number of non-identity mappings between residual blocks, and allows an identity mapping directly from the input to the output. This design was used to train models with 200 to over 1000 layers, and was found to consistently outperform variants where the residual path is not an identity function. The pre-activation ResNet with 200 layers took 3 weeks to train for ImageNet on 8 GPUs in 2016. Since GPT-2, transformer blocks have been mostly implemented as pre-activation blocks. This is often referred to as "pre-normalization" in the literature of transformer models.

>> Section 2

Pre-activation block The pre-activation residual block applies activation functions before applying the residual function F . Formally, the computation of a pre-activation residual block can be written as:

>> Section 3

Forward propagation If the output of the ℓ -th residual block is the input to the $(\ell + 1)$ -th residual block (assuming no activation function between blocks), then the $(\ell + 1)$ -th input is:

>> Section 4

This formulation suggests that the gradient computation of a shallower layer, $\frac{\partial E}{\partial x_{\ell}}$, always has a later term $\frac{\partial E}{\partial x_L}$ that is directly added. Even if the gradients of the $F(x_i)$ terms are small, the total gradient $\frac{\partial E}{\partial x_{\ell}}$ resists vanishing due to the added term $\frac{\partial E}{\partial x_L}$.

>> Section 5

Applications Originally, ResNet was designed for computer vision. All transformer architectures include residual connections. Indeed, very deep transformers cannot be trained without them. The original ResNet paper made no claim on being inspired by biological systems. However, later research has related ResNet to biologically-plausible

algorithms. A study published in Science in 2023 disclosed the complete connectome of an insect brain (specifically that of a fruit fly larva). This study discovered "multilayer shortcuts" that resemble the skip connections in artificial neural networks, including ResNets.

>> Section 6

Previous work Residual connections were noticed in neuroanatomy, such as Lorente de No (1938). McCulloch and Pitts (1943) proposed artificial neural networks and considered those with residual connections. In 1961, Frank Rosenblatt described a three-layer multilayer perceptron (MLP) model with skip connections. The model was referred to as a "cross-coupled system", and the skip connections were forms of cross-coupled connections.

>> Section 7

Subsequent work Wide Residual Network (2016) found that using more channels and fewer layers than the original ResNet improves performance and GPU-computational efficiency, and that a block with two 3×3 convolutions is superior to other configurations of convolution blocks. DenseNet (2016) connects the output of each layer to the input to each subsequent layer:

>> Section 8

Degradation problem Sepp Hochreiter discovered the vanishing gradient problem in 1991 and argued that it explained why the then-prevalent forms of recurrent neural networks did not work for long sequences. He and Schmidhuber later designed the LSTM architecture to solve this problem, which has a "cell state" c_t that can function as a generalized residual connection. The highway network (2015) applied the idea of an LSTM unfolded in time to feedforward neural networks, resulting in the highway network. ResNet is equivalent to an open-gated highway network.

>> Section 9

Residual connection In a multilayer neural network model, consider a (non-residual) subnetwork with a certain number of stacked layers (e.g., 2 or 3). Let $H(x; \alpha)$ denote the subnetwork. Suppose H^* is the desired optimal output of this subnetwork. Residual learning simply adds x directly to the output, such that the optimal learned output now becomes $H^* - x$, which is interpreted as a "residual" with respect to x . The operation of "adding x " is implemented via a "skip connection" that performs an identity mapping to connect the input of the subnetwork with its output. This connection is referred to as a "residual connection" in later work. Let $F(x; \alpha) =$

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$H(x; \alpha) = x + F(x; \alpha)$. The function F is often represented by matrix multiplication interlaced with activation functions and normalization operations (e.g., batch normalization or layer normalization). As a whole, one of these subnetworks is referred to as a "residual block". A deep residual network is constructed by simply stacking these blocks. Long short-term memory (LSTM) has a memory mechanism that serves as a residual connection. In an LSTM without a forget gate, an input x_t is processed by a function F and added to a memory cell c_t , resulting in $c_{t+1} = c_t + F(x_t)$. An LSTM with a forget gate essentially functions as a highway network. To stabilize the variance of the layers' inputs, it is recommended to replace the residual connections $x + f(x)$ with $x/L + f(x)$, where L is the total number of residual layers.

>> Section 10

During the early days of deep learning, there were attempts to train increasingly deep models. Notable examples included the AlexNet (2012), which had 8 layers, and the VGG-19 (2014), which had 19 layers. However, stacking too many layers led to a steep reduction in training accuracy, known as the "degradation" problem. In theory, adding additional layers to deepen a network should not result in a higher training loss, but this is what happened with VGGNet. If the extra layers can be set as identity mappings, however, then the deeper network would represent the same function as its shallower counterpart. There is some evidence that the optimizer is not able to approach identity mappings for the parameterized layers, and the benefit of residual connections was to allow identity mappings by default. In 2014, the state of the art was training deep neural networks with 20 to 30 layers. The research team for ResNet attempted to train deeper ones by empirically testing various methods for training deeper networks, until they came upon the ResNet architecture.